12.359

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Question

For real numbers a, b, c, let $\mathbf{M} = \begin{pmatrix} a & ac & 0 \\ 1 & c & 0 \\ b & bc & 1 \end{pmatrix}$. Then, which of the following statements is TRUE?

1. $\text{Rank}(\mathbf{M}) = 3$ for every $a, b, c \in \mathbb{R}$
2. If $a + c = 0$ then $\mathbf{M}$ is diagonalizable for every $b \in \mathbb{R}$
3. $\mathbf{M}$ has a pair of orthogonal eigenvectors for every $a, b, c \in \mathbb{R}$
4. If $b = 0$ and $a + c = 1$ then M is NOT idempotent

Theoretical Solution

First, let's find the eigenvalues of $\mathbf{M}$ by solving the characteristic equation $\left|\mathbf{M} - \lambda\mathbf{I}\right| = 0$.

$$\begin{vmatrix} a-\lambda & ac & 0 \\ 1 & c-\lambda & 0 \\ b & bc & 1-\lambda \end{vmatrix} = 0 \tag{1}$$

Expanding along the third column:

$$\begin{aligned} (1-\lambda)\begin{vmatrix} a-\lambda & ac \\ 1 & c-\lambda \end{vmatrix} &= 0 \\ (1-\lambda)[(a-\lambda)(c-\lambda) - ac] &= 0 \\ (1-\lambda)[ac - a\lambda - c\lambda + \lambda^2 - ac] &= 0 \\ (1-\lambda)[\lambda^2 - (a+c)\lambda] &= 0 \\ (1-\lambda)\lambda(\lambda - (a+c)) &= 0 \end{aligned}$$

The eigenvalues are $\lambda_1 = 1$, $\lambda_2 = 0$, and $\lambda_3 = a + c$.

Theoretical Solution

1) Rank(M) $= 3$ for every $a, b, c \in \mathbb{R}$
The rank of a square matrix is equal to its dimension if and only if its determinant is non-zero. Let's calculate the determinant of M.

$$\left|\mathbf{M}\right| = \begin{vmatrix} a & ac & 0 \\ 1 & c & 0 \\ b & bc & 1 \end{vmatrix} \tag{2}$$

Expanding along the third column:

$$\left|\mathbf{M}\right| = 1 \cdot \begin{vmatrix} a & ac \\ 1 & c \end{vmatrix} = a(c) - ac(1) = ac - ac = 0 \tag{3}$$

Since the determinant of M is always 0, its rank can never be 3.
Therefore, statement 1 is FALSE.

2) If $a + c = 0$ then M is diagonalizable for every $b \in \mathbb{R}$

If $a + c = 0$, the eigenvalues of M are $\lambda = 1, 0, 0$. For the matrix to be diagonalizable, the algebraic multiplicity (AM) of each eigenvalue must equal its geometric multiplicity (GM). The eigenvalue $\lambda = 0$ has an AM of 2. Its GM is given by $n - \text{rank}(\mathbf{M} - 0I) = 3 - \text{rank}(\mathbf{M})$. We need the GM to be 2, which means we need $\text{rank}(\mathbf{M}) = 1$.

Let's check the rank of M when $a + c = 0$ (i.e., $c = -a$).

$$\mathbf{M} = \begin{pmatrix} a & -a^2 & 0 \\ 1 & -a & 0 \\ b & -ab & 1 \end{pmatrix} \tag{4}$$

Theoretical Solution

Consider the 2x2 minor formed by rows 2 and 3, and columns 2 and 3:

$$\begin{vmatrix} -a & 0 \\ -ab & 1 \end{vmatrix} = -a \tag{5}$$

If $a \neq 0$, this minor is non-zero, which means the rank of M is at least 2. Since $\left|\mathbf{M}\right| = 0$, the rank is exactly 2. If $\text{rank}(\mathbf{M}) = 2$, the GM of $\lambda = 0$ is $3 - 2 = 1$. This does not equal the AM of 2. Thus, the matrix is not diagonalizable when $a \neq 0$.

Therefore, statement 2 is FALSE.

3) M has a pair of orthogonal eigenvectors for every $a, b, c \in \mathbb{R}$
Let's find the eigenvectors for two distinct eigenvalues, $\lambda_1 = 1$ and $\lambda_2 = 0$.
Eigenvector for $\lambda_1 = 1$: We solve $(\mathbf{M} - I)v = 0$:

$$\begin{pmatrix} a-1 & ac & 0 \\ 1 & c-1 & 0 \\ b & bc & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \tag{6}$$

The third column is all zeros, which implies that a vector of the form $v_1 = (0, 0, 1)^T$ is a valid eigenvector.
Eigenvector for $\lambda_2 = 0$: We solve $\mathbf{M}v = 0$:

$$\begin{pmatrix} a & ac & 0 \\ 1 & c & 0 \\ b & bc & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \tag{7}$$

Theoretical Solution

From the first two rows, we get $x + cy = 0$. A possible non-trivial solution is $x = -c, y = 1$. Substituting into the third row:
$b(-c) + bc(1) + z = 0 \implies z = 0$. So, an eigenvector is $v_2 = (-c, 1, 0)^T$.
Orthogonality Check:

$$v_1 \cdot v_2 = (0, 0, 1) \cdot (-c, 1, 0) = 0 \cdot (-c) + 0 \cdot 1 + 1 \cdot 0 = 0 \tag{8}$$

Since their dot product is 0, the eigenvectors are orthogonal. This pair exists for any $a, b, c \in \mathbb{R}$.
Therefore, statement 3 is TRUE.

Theoretical Solution

4)If $b = 0$ and $a + c = 1$ then $\mathbf{M}$ is NOT idempotent
An idempotent matrix is one for which $\mathbf{M}^2 = \mathbf{M}$. Applying the conditions $b = 0$ and $a + c = 1$:

$$\mathbf{M} = \begin{pmatrix} a & ac & 0 \\ 1 & c & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{9}$$

Now we compute $\mathbf{M}^2$:

$$\mathbf{M}^2 = \begin{pmatrix} a & ac & 0 \\ 1 & c & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & ac & 0 \\ 1 & c & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a^2 + ac & a^2c + ac^2 & 0 \\ a + c & ac + c^2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{10}$$

Theoretical Solution

Using the condition $a + c = 1$:

- $a^2 + ac = a(a + c) = a(1) = a$
- $a^2c + ac^2 = ac(a + c) = ac(1) = ac$
- $a + c = 1$
- $ac + c^2 = c(a + c) = c(1) = c$

Substituting these back into $\mathbf{M}^2$:

$$\mathbf{M}^2 = \begin{pmatrix} a & ac & 0 \\ 1 & c & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{M} \tag{11}$$

Since $\mathbf{M}^2 = \mathbf{M}$, the matrix is idempotent. The statement says it is NOT idempotent.

Therefore, statement 4 is FALSE.